



THE KING'S SCHOOL

2011 Higher School Certificate Trial Examination

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question
- Answer each question in a separate booklet

Total marks – 120

- Attempt Questions 1-8
- All questions are of equal value

Disclaimer

This is a Trial Higher School Certificate Examination only. Whilst it reflects and mirrors both the format and topics of the Higher School Certificate Examination designed by the NSW Board of Studies for the respective sections, there is no guarantee that the content of this examination exactly replicates the actual Higher School Certificate Examination.

Total marks – 120
Attempt Questions 1-8
All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Question 1 (15 marks) Use a SEPARATE writing booklet **Marks**

(a) (i) Use the table of standard integrals to show that

$$\int_0^1 \frac{dx}{\sqrt{4+x^2}} = \ln \left(\frac{1+\sqrt{5}}{2} \right) \quad 2$$

(ii) Evaluate $\int_0^{\frac{\pi}{4}} \frac{\sec^2 x}{\sqrt{4+\tan^2 x}} dx$ **2**

(b) Find $\int \frac{2x+3}{x+1} dx$ **2**

(c) By using the substitution $t = \tan \frac{\theta}{2}$, or otherwise, evaluate

$$\int_0^{\frac{\pi}{2}} \frac{1}{2+\cos \theta} d\theta \quad 3$$

(d) Let $u_n = \int_1^e x (\ln x)^n dx$, $n = 0, 1, 2, \dots$

(i) Use integration by parts to show that

$$2u_n = e^2 - n u_{n-1}, \quad n = 1, 2, 3, \dots \quad 4$$

(ii) Hence, or otherwise, evaluate $\int_1^e x (\ln x)^2 dx$ **2**

End of Question 1

(a) Let $z = 1 - i$ and $w = -1 + 2i$

(i) Find $|zw|$ **2**

(ii) Show that $z + iw = \bar{w}$ **2**

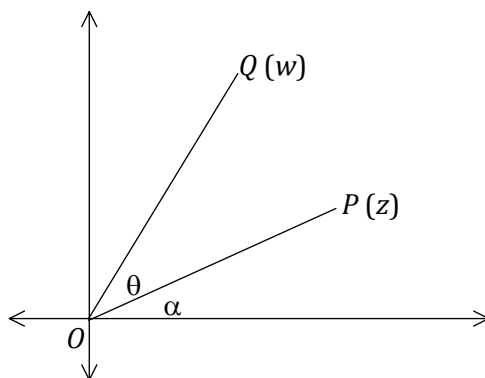
(iii) Find $\arg\left(\frac{12}{z + w}\right)$ **2**

(b) Let $z = \frac{\sqrt{3}}{2} + \frac{1}{2}i$

(i) Express z in modulus-argument form. **1**

(ii) Simplify $1 + z^2 + z^4 + \dots + z^{20} + z^{22}$ **3**

(c) Let $z = \cos \alpha + i \sin \alpha$, $0 < \alpha < \frac{\pi}{4}$



On the Argand diagram the point P represents the complex number z and Q represents the complex number w .

$$OP = OQ \text{ and } \angle QOP = \theta, 0 < \theta \leq \frac{\pi}{4}$$

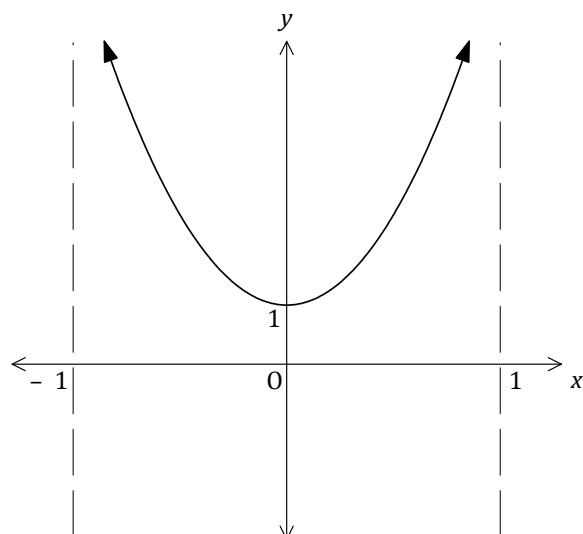
(i) Write down the complex number w . **1**

(ii) Show that $z\bar{w} = \cos \theta - i \sin \theta$ **2**

(iii) By considering $\triangle OPQ$, or otherwise, deduce that $\cos \frac{\theta}{2} = -\frac{\operatorname{Im}(z\bar{w})}{|z - w|}$ **2**

End of Question 2

(a)



The diagram shows the sketch of $f(x) = \frac{1}{\sqrt{1-x^2}}$.

Draw sketches of the graphs of the following:

(i) $y = f^{-1}(x), \quad y \leq 0$

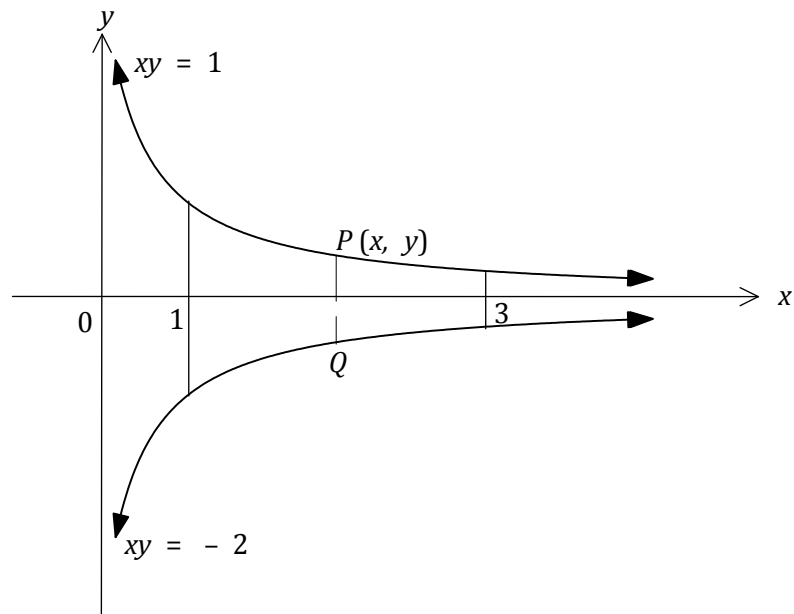
1

(ii) $y = \frac{x}{\sqrt{x^2 - x^4}}$

2

Question 3 continues on the next page

(b)



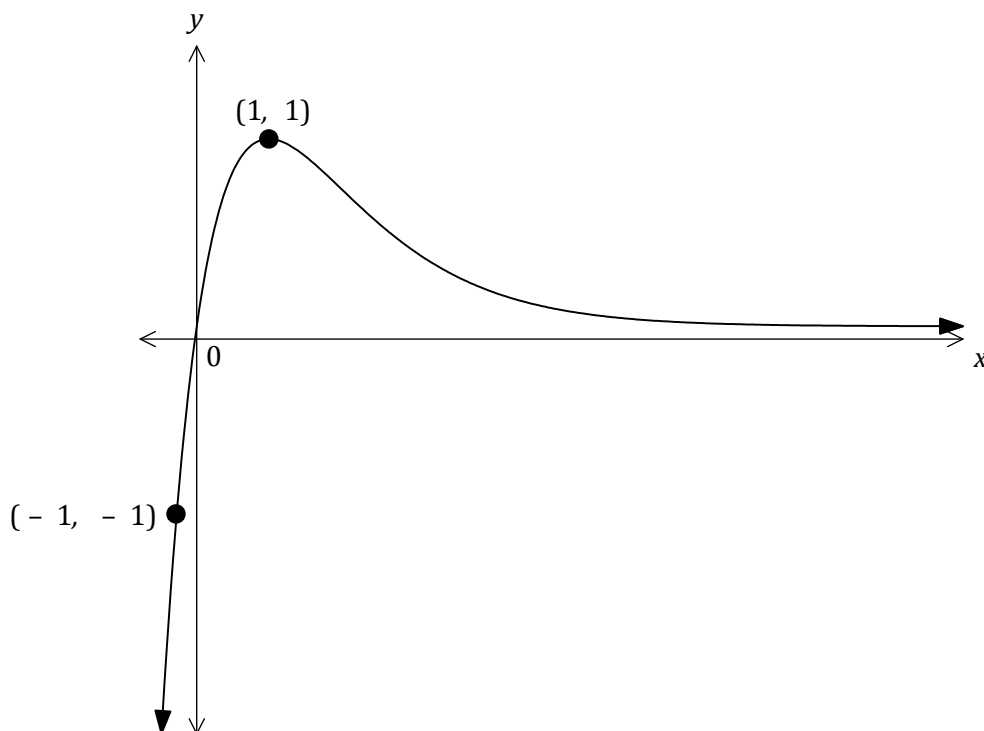
The enclosed region between $xy = 1$ and $xy = -2$ from $x = 1$ to $x = 3$ is the base of a solid.

Cross-sections of the solid perpendicular to this base and the x axis are equilateral triangles.

- (i) Show that the area of the smallest cross-section is $\frac{\sqrt{3}}{4}$ 1
- (ii) Find the volume of the solid. 3

Question 3 continues on the next page

(c)



The sketch shows the graph of $y = f(x)$.

Draw separate sketches of the following:

(i) $y = f'(x)$ 2

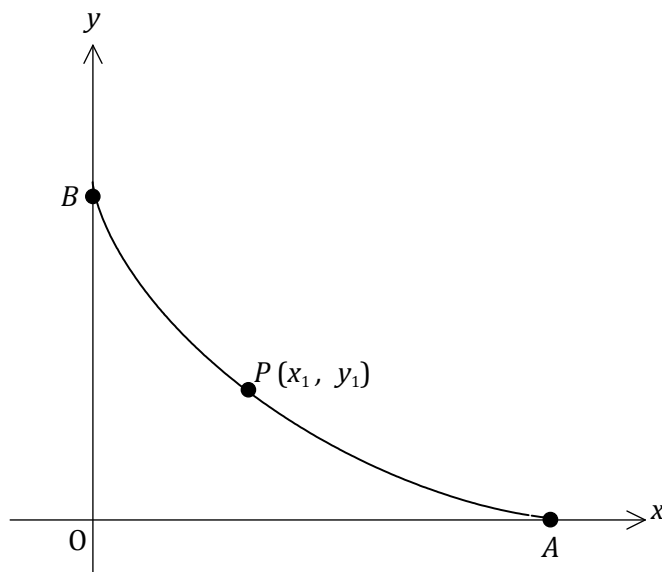
(ii) $y = \ln f(x)$ 2

(iii) $y = f(x + |x|)$ 2

(iv) $y = \cos^{-1} f(x)$ 2

End of Question 3

(a)



The sketch shows the graph of $x^{\frac{2}{3}} + y^{\frac{2}{3}} = 4$, $x \geq 0$, $y \geq 0$, meeting the axes at A , B

(i) Find the coordinates of A 1

(ii) Show that $\frac{dy}{dx} = -\left(\frac{y}{x}\right)^{\frac{1}{3}}$, $x \neq 0$ 2

(iii) Prove that the equation of the tangent at $P(x_1, y_1)$ is

$$y = -\left(\frac{y_1}{x_1}\right)^{\frac{1}{3}}x + 4y_1^{\frac{1}{3}}, \quad x_1 \neq 0$$
2

(iv) The tangent at $P(x_1, y_1)$ meets the x axis at X and the y axis at Y .

Find the length of XY . 2

Question 4 continues on the next page

(b) Suppose $\alpha, \beta, \gamma, \delta$ are the four roots of the polynomial equation

$$P(x) = x^4 - Ax - 1 = 0 \text{ where } A \text{ is real.}$$

(i) By considering $P''(x)$, or otherwise, show that at most two of the roots are real.

3

(ii) Prove that the polynomial equation with the four roots $\alpha^2, \beta^2, \gamma^2, \delta^2$ is $x^4 - 2x^2 - A^2x + 1 = 0$

3

(iii) By considering another suitable polynomial equation, or otherwise, prove that

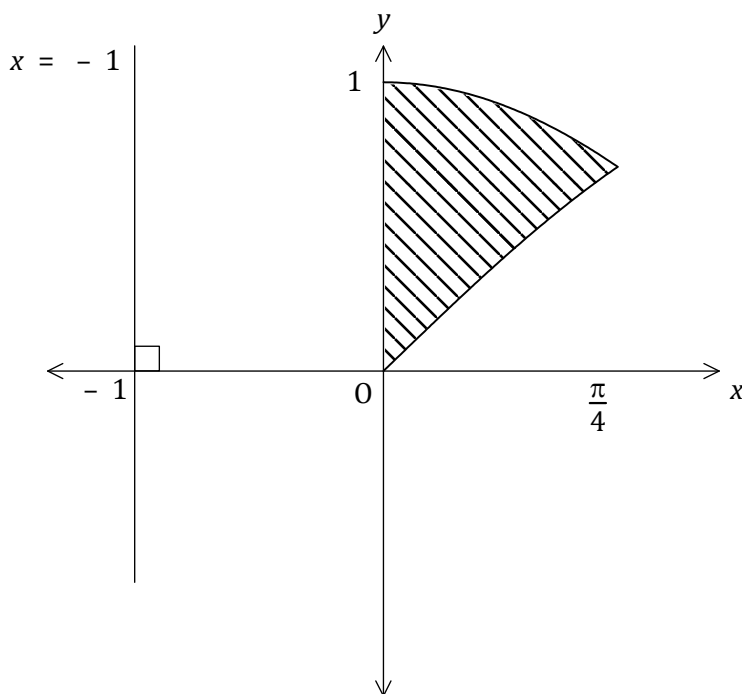
$$(\alpha^2 + 1)(\beta^2 + 1)(\gamma^2 + 1)(\delta^2 + 1) = A^2$$

2

End of Question 4

- (a) (i) Find the derivative of $x(\sin x + \cos x) + \cos x - \sin x$ **1**

(ii)



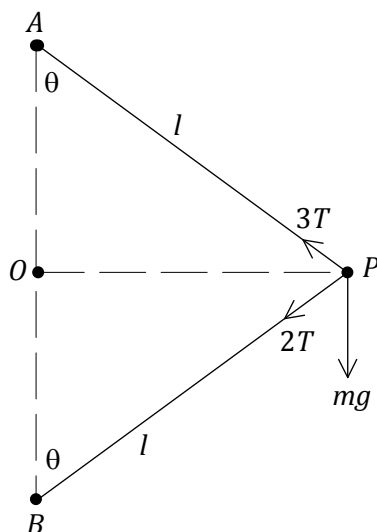
The diagram shows the region bounded by the curves $y = \sin x$ and $y = \cos x$ and the y axis.

The region is revolved about the line $x = -1$

Use the method of cylindrical shells to find the volume of the solid of revolution. **4**

Question 5 continues on the next page

(b)



The ends of a string of length $2l$ are attached to two points A and B where B is vertically below A .

A particle P of mass m is attached at the mid-point of the string and rotates about O , vertically below A , at constant angular speed ω .

The tensions in the strings PA and PB are $3T$ and $2T$, respectively.

g is the acceleration due to gravity.

Let $\angle PAO = \theta$

(i) Show that $T \cos \theta = mg$ 2

(ii) Show that $m\omega^2 l = 5T$ 2

(iii) Prove that $\omega > \sqrt{\frac{5g}{l}}$ 2

(c) S and S' are the two foci of the ellipse $\frac{x^2}{9} + \frac{y^2}{5} = 1$

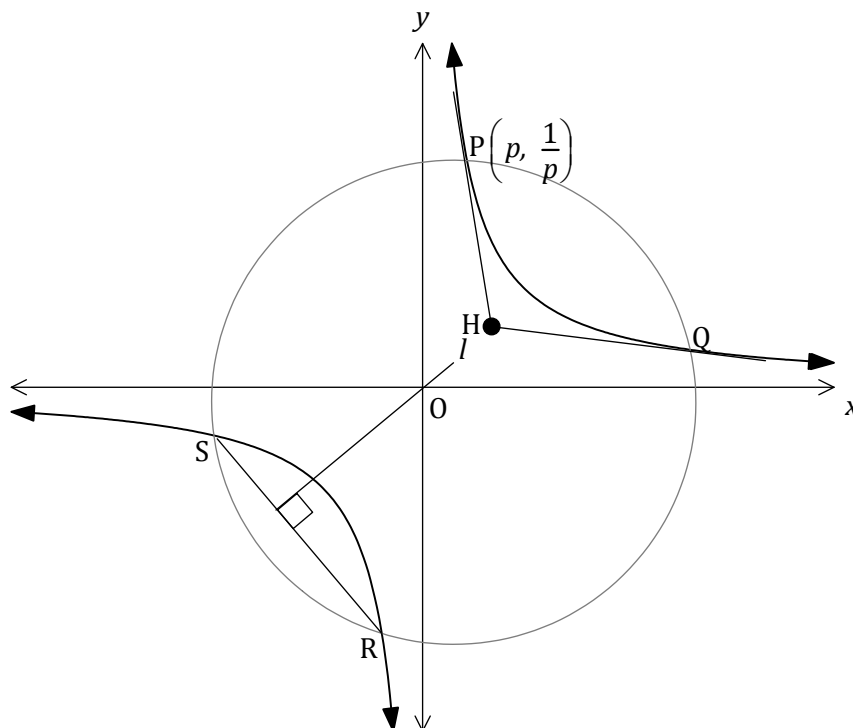
e is the eccentricity of the ellipse.

(i) Find S and S' 1

(ii) $P(x_1, y_1)$ is any point on the ellipse. Prove that $PS + PS' = 9e$ 3

End of Question 5

(a)



The diagram shows the hyperbola $xy = 1$ and the circle $x^2 + y^2 + Ax + By + C = 0$ meeting at the points P, Q, R, S.

Let $x = t$, $y = \frac{1}{t}$ be the parametric equations of the hyperbola.

The parameters at P, Q, R, S are p, q, r, s respectively so that $P = \left(p, \frac{1}{p}\right)$

- (i) Prove that $pqrs = 1$ 2
- (ii) Show that the equation of the line l passing through O (0, 0) and perpendicular to RS is $y = rsx$ 2
- (iii) Prove that the equation of the tangent at $P\left(p, \frac{1}{p}\right)$ on the hyperbola is $x + p^2y = 2p$ 2
- (iv) Prove that the line l passes through the point of intersection H of the tangents at P and Q. 3

Question 6 continues on the next page

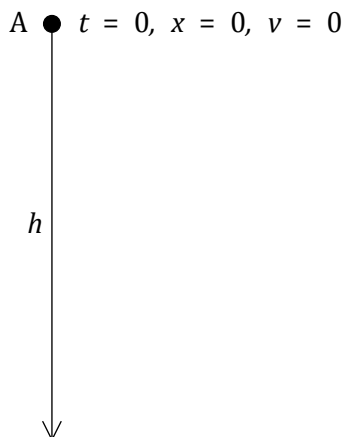
(b) (i) Express $\frac{7n + 3}{n(n + 1)(n + 3)}$ in the form $\frac{a}{n} + \frac{b}{n + 1} + \frac{c}{n + 3}$ **3**

(ii) Let $S_n = \sum_1^n \left(\frac{a}{n} + \frac{b}{n + 1} + \frac{c}{n + 3} \right) = \sum_1^n \frac{a}{n} + \sum_1^n \frac{b}{n + 1} + \sum_1^n \frac{c}{n + 3}$

Prove that $S_n < \frac{7}{2}$ **3**

End of Question 6

(a)

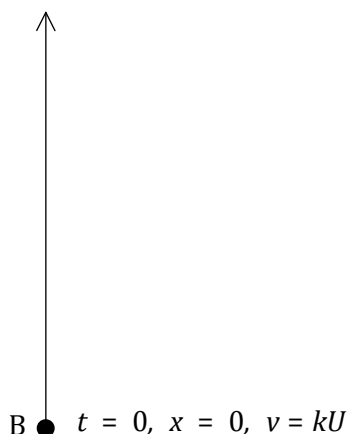


A particle A of mass m falls vertically from rest at a height h above the ground. It is subject to a resistance mkv where v is its speed and k is a positive constant. g is the acceleration due to gravity.

- (i) Using the vertical axis as in the diagram, show that $\ddot{x} = k(U - v)$ where U is the terminal speed of the particle.

2

(ii)



At the same time as particle A falls from rest another particle B of mass m is projected upward in the same vertical line as particle A with a speed of kU , where U is the terminal speed of the particle.

Using the vertical axis as in the diagram, show that

$$\frac{dv}{dx} = \frac{-k(U + v)}{v}$$

1

Question 7 continues on the next page

- (iii) Prove that the greatest possible height reached by particle B is

$$H = U \left(1 - \frac{1}{k} \ln(1 + k) \right) \quad 3$$

- (iv) Prove by using the equation of motion in (i) that the speed of particle A at any time t is given by $v = U(1 - e^{-kt})$ 3

- (v) Similarly, it can be shown that for particle B, $v = kUe^{-kt} - U(1 - e^{-kt})$

[DO NOT PROVE THIS]

The two particles collide after time T when particle A has fallen a distance X .

Given that $h < H$, prove that $T = \frac{1}{k} \ln \left(\frac{U}{U - h} \right)$ 3

- (b) A point moves in the xy plane so that its coordinates at any time $t \geq 0$ are given by $x = 1 + \sin t$, $y = -\cos 2t$

Show that the point moves on a parabola and sketch the path of its motion. 3

End of Question 7

(a) Let $J_n = \int_2^4 \frac{dx}{x \sqrt{x^{2n} - 1}}$, $n = 1, 2, 3, \dots$

(i) Show that $J_n < \frac{\ln 2}{\sqrt{2^{2n} - 1}}$ **2**

(ii) Show that $J_n > \frac{2^n - 1}{n \cdot 4^n}$ **2**

(iii) Show that $\frac{1}{x \sqrt{x^{2n} - 1}} = \frac{x^{-n-1}}{\sqrt{1 - x^{-2n}}}$ **1**

(iv) Find J_n **3**

(b) A sequence $u_1, u_2, u_3, \dots$ is given by

$$u_1 = 0, u_2 = 1 \text{ and } u_n = (n - 1)(u_{n-1} + u_{n-2}), n \geq 3$$

(i) Find u_4 **1**

(ii) Let $A_n = \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \dots + (-1)^n \frac{1}{n!}$

Show that $A_{n-2} - A_{n-1} = (-1)^n \frac{1}{(n-1)!}$ **1**

(iii) Prove by induction for $n \geq 3$ that $u_n = n! A_n$ **5**

End of Examination Paper

--	--	--	--	--	--	--	--	--

Student Number



THE KING'S SCHOOL

2011 Higher School Certificate Trial Examination

Mathematics Extension 2

Question	(Marks)	Complex Numbers	Functions	Harder Extension 1	Integration	Conics	Mechanics
1	(15)				$a-d$ /15		
2	(15)	$a-c$ /15					
3	(15)		a, c /11		b /14		
4	(15)		a, b /15				
5	(15)				a / 5	c / 4	b / 6
6	(15)		b / 6			a / 9	
7	(15)		b / 3				a /12
8	(15)			b / 7	a / 8		
Total	(120)	/15	/35	/ 7	/32	/13	/18

Question 1

$$(a) \quad (i) \quad I = \left[\ln(x + \sqrt{x^2 + 4}) \right]_0^1$$

$$= \ln(1 + \sqrt{5}) - \ln 2 = \ln\left(\frac{1 + \sqrt{5}}{2}\right)$$

$$(ii) \quad \text{Put } u = \tan x \quad x=0, u=0$$

$$\frac{du}{dx} = \sec^2 x \quad x = \frac{\pi}{4}, u=1$$

$$\therefore I = \int_0^1 \frac{du}{\sqrt{4+u^2}} = \ln\left(\frac{1+\sqrt{5}}{2}\right)$$

$$(b) \quad I = \int \frac{2(x+1) + 1}{x+1} dx = \int 2 + \frac{1}{x+1} dx = 2x + \ln(x+1) + C$$

$$(c) \quad t = \tan \frac{\theta}{2} \quad \theta=0, t=0$$

$$\frac{dt}{d\theta} = \frac{1}{2} \sec^2 \frac{\theta}{2} = \frac{1}{2} (1+t^2) \quad \theta = \frac{\pi}{2}, t=1$$

$$\therefore I = \int_0^1 \frac{2 dt}{2(1+t^2) + 1 - t^2}$$

$$= \int_0^1 \frac{2 dt}{3 + t^2}$$

$$= \frac{2}{\sqrt{3}} \left[\tan^{-1} \frac{t}{\sqrt{3}} \right]_0^1$$

$$= \frac{2}{\sqrt{3}} \left(\frac{\pi}{6} - 0 \right) = \frac{\pi}{3\sqrt{3}} \quad \text{or} \quad \frac{\sqrt{3}\pi}{9}$$

$$(d) \quad (i) \quad u = (\ln x)^n, \quad \frac{dv}{dx} = x$$

$$\frac{du}{dx} = n \frac{(\ln x)^{n-1}}{x}, \quad v = \frac{x^2}{2}$$

$$\therefore u_n = \left[\frac{x^2}{2} (\ln x)^n \right]_1^e - \frac{n}{2} \int_1^e x (\ln x)^{n-1} dx$$

$$= \frac{e^2}{2} - \frac{n}{2} u_{n-1}$$

$$\text{i.e. } 2u_n = e^2 - n u_{n-1}$$

$$\begin{aligned} (ii) \quad 2u_2 &= e^2 - 2u_1 \\ &= e^2 - (e^2 - u_0) \\ &= u_0 = \int_1^e x \, dx = \left[\frac{x^2}{2} \right]_1^e = \frac{e^2}{2} - \frac{1}{2} \end{aligned}$$

$$\Rightarrow \int_1^e x (\ln x)^2 \, dx = \frac{1}{4} (e^2 - 1)$$

Question 2

(a) (i) $|zw| = |z||w| = \sqrt{2}\sqrt{5} = \sqrt{10}$

(ii) $z + iw = 1 - i + i(-1 + 2i)$
 $= 1 - i - i - 2 = -1 - 2i = \bar{w}$

(iii) $z + w = i$

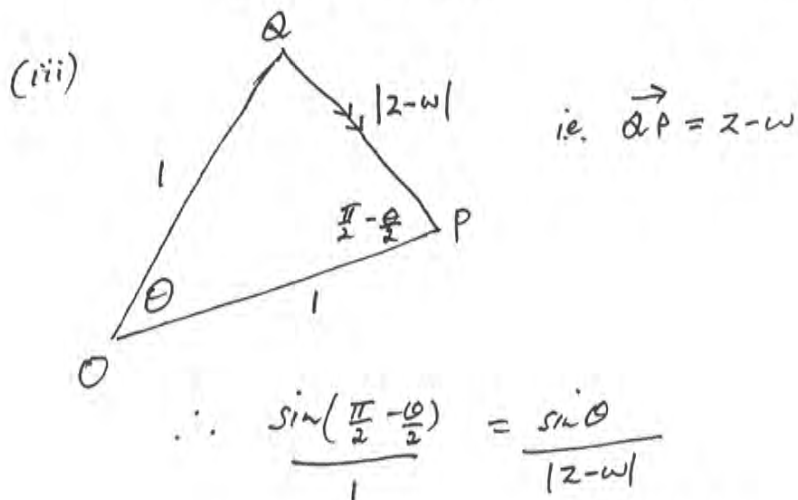
$\therefore \arg\left(\frac{12}{i}\right) = \arg 12 - \arg i = -\frac{\pi}{2}$

(b) (i) $z = \cos \frac{\pi}{6} + i \sin \frac{\pi}{6}$

(ii) $1 + z^2 + \dots + z^{22} = \frac{(z^2)^{12} - 1}{z^2 - 1} = \frac{z^{24} - 1}{z^2 - 1} = \frac{\cos 4\pi + i \sin 4\pi - 1}{z^2 - 1}$
 $= 0$

(c) (i) $w = \cos(\alpha + \theta) + i \sin(\alpha + \theta)$

(ii) $z\bar{w} = (\cos \alpha + i \sin \alpha)(\cos(\alpha + \theta) - i \sin(\alpha + \theta))$
 $= \cos \alpha \cos(\alpha + \theta) + \sin \alpha \sin(\alpha + \theta) + i(\sin \alpha \cos(\alpha + \theta) - \cos \alpha \sin(\alpha + \theta))$
 $= \cos(\alpha + \theta - \alpha) - i \sin(\alpha + \theta - \alpha)$
 $= \cos \theta - i \sin \theta$ [ALTERNATIVES]

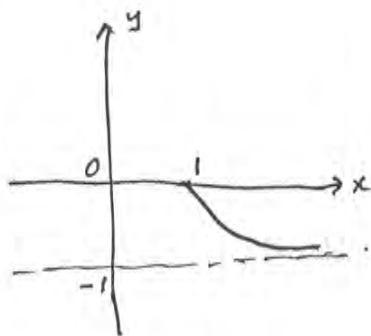


$\therefore \frac{\sin\left(\frac{\pi}{2} - \frac{\theta}{2}\right)}{1} = \frac{\sin \theta}{|z-w|}$

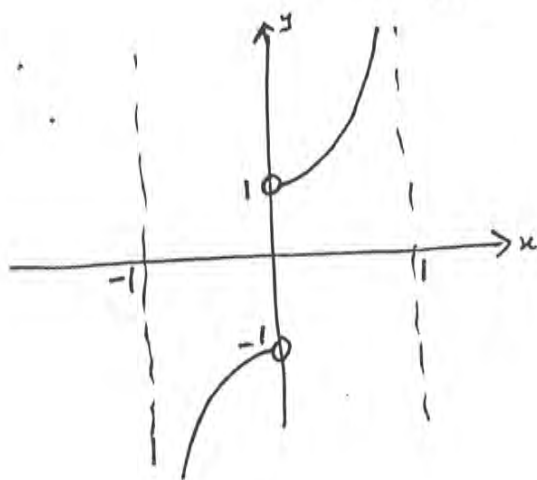
ie. $\cos \frac{\theta}{2} = \frac{-\operatorname{Im}(z\bar{w})}{|z-w|}$, from (ii)

Question 3

(a) (i)



$$(ii) \quad y = \frac{x}{\sqrt{x^2} \sqrt{1-x^2}} = \frac{x}{x \sqrt{1-x^2}} \quad \text{if } x > 0$$
$$= \frac{1}{\sqrt{1-x^2}}$$
$$\& \quad y = -\frac{1}{\sqrt{1-x^2}} \quad \text{if } x < 0$$



(b) (i) least occurs at $x=3$

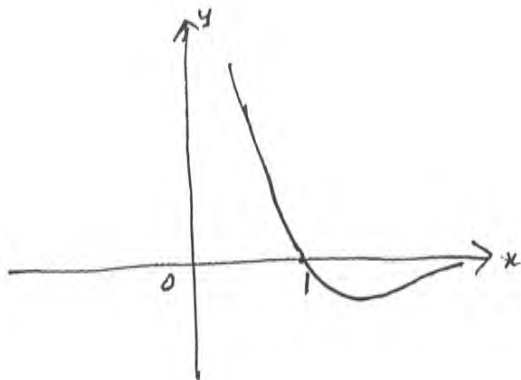
$$\therefore \text{Area} = \frac{1}{2} \left(\frac{1}{3} + \frac{2}{3} \right)^2 \sin 60^\circ = \frac{\sqrt{3}}{4}$$

$$(ii) \quad PQ = \frac{1}{x} + \frac{2}{x} = \frac{3}{x}$$

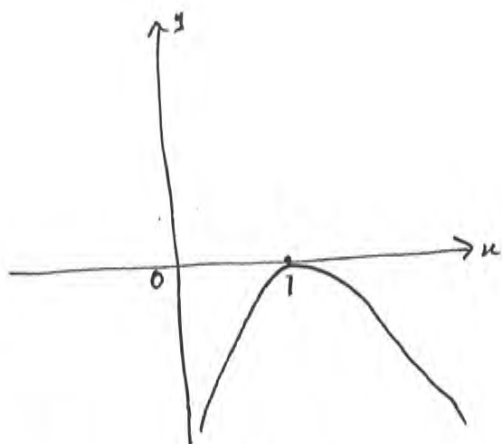
$$\therefore \text{Area cross-section} = \frac{1}{2} \cdot \frac{9}{x^2} \cdot \frac{\sqrt{3}}{2} = \frac{9\sqrt{3}}{4x^2}$$

$$\therefore \text{Volume} = \frac{9\sqrt{3}}{4} \int_1^3 \frac{1}{x^2} dx = -\frac{9\sqrt{3}}{4} \left[\frac{1}{x} \right]_1^3$$
$$= -\frac{9\sqrt{3}}{4} \left(\frac{1}{3} - 1 \right)$$
$$= \frac{3\sqrt{3}}{2}$$

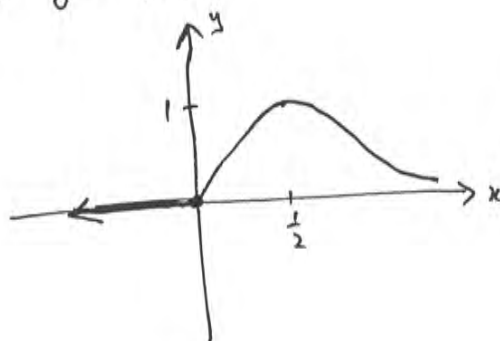
(c) (i)



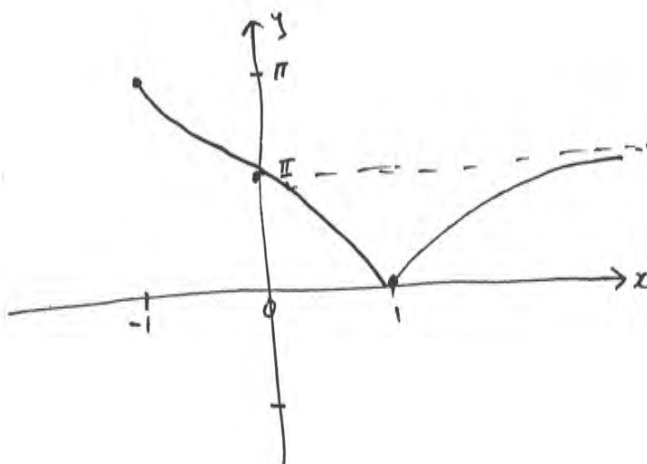
(ii)



(iii) If $x \geq 0$, $y = f(2x)$
 If $x \leq 0$, $y = f(0)$



(iv)



Question 4

(a) (i) at A, $y=0 \quad \therefore x = 4^{3/2} = 8 \quad \text{i.e. } A = (8, 0)$

(ii) $\frac{2}{3} x^{-\frac{1}{3}} + \frac{2}{3} y^{-\frac{1}{3}} \frac{dy}{dx} = 0$

$$\therefore \frac{dy}{dx} = -\frac{x^{-\frac{1}{3}}}{y^{-\frac{1}{3}}} = -\left(\frac{y}{x}\right)^{\frac{1}{3}}$$

(iii) Tangent is $y - y_1 = -\left(\frac{y_1}{x_1}\right)^{\frac{1}{3}} (x - x_1)$

$$\text{i.e. } y = -\left(\frac{y_1}{x_1}\right)^{\frac{1}{3}} x + y_1^{\frac{1}{3}} x_1^{\frac{2}{3}} + y_1$$

$$= -\left(\frac{y_1}{x_1}\right)^{\frac{1}{3}} x + y_1^{\frac{1}{3}} (x_1^{1/3} + y_1^{1/3})$$

$$\text{i.e. } y = -\left(\frac{y_1}{x_1}\right)^{\frac{1}{3}} x + 4y_1^{\frac{1}{3}}$$

(iv) At Y, $x=0 \rightarrow y = 4y_1^{\frac{1}{3}}$

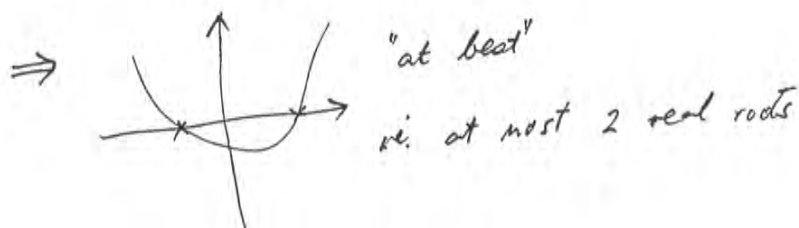
$\therefore$, on symmetry, at X, $x = 4x_1^{\frac{1}{3}}$

$$\therefore XY = 4\sqrt{x_1^{1/3} + y_1^{1/3}} = 4\sqrt{4} = 8$$

(b) (i) $p'(x) = 4x^3 - A$

$$p''(x) = 12x^2 \geq 0 \quad \forall x$$

$\Rightarrow$ curve is concave upward $\forall x$



$$(ii) \text{ Put } x = d^2 \Rightarrow d = \sqrt{x}$$

$$\therefore \text{ poly is } (\sqrt{x})^4 - A\sqrt{x} - 1 = 0$$

$$\text{or } A\sqrt{x} = x^2 - 1$$

$$\therefore A^2 x = x^4 - 2x^2 + 1$$

$$\text{i.e. } x^4 - 2x^2 - A^2 x + 1 = 0$$

$$(iii) \text{ Put } x = d^2 + 1 \Rightarrow d^2 = x - 1$$

$\therefore$ from (ii), eqn with roots $d^2+1, \dots, f^2+1$ is

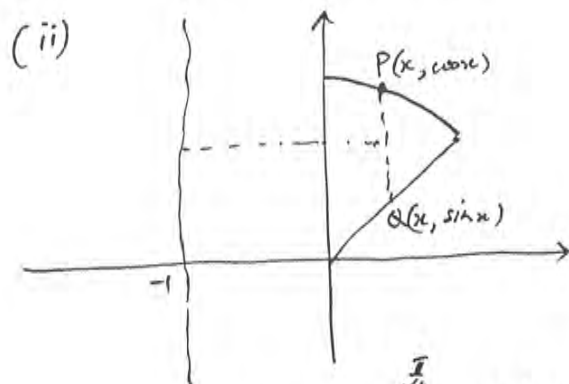
$$(x-1)^4 - 2(x-1)^2 - A^2(x-1) + 1 = 0$$

$$\& (d^2+1)(f^2+1)(g^2+1)(h^2+1) \text{ is the product of the roots}$$

$$= 1 - 2 + A^2 + 1 = A^2$$

Question 5

(a) (i) $x(\cos x - \sin x) + \sin x + \cos x - \sin x - \cos x = x(\cos x - \sin x)$



$$\therefore \delta V \approx \pi [(x+\delta x+1)^2 - (x+1)^2] (\cos x - \sin x)$$

$$\approx \pi (2(x+1)\delta x) (\cos x - \sin x)$$

$$\therefore V = 2\pi \int_0^{\frac{\pi}{4}} (x+1)(\cos x - \sin x) dx$$

$$= 2\pi \int_0^{\frac{\pi}{4}} x(\cos x - \sin x) + \cos x - \sin x dx$$

$$= 2\pi \left[x(\sin x + \cos x) + \cos x - \sin x + \sin x + \cos x \right]_0^{\frac{\pi}{4}} \text{ from (i)}$$

$$= 2\pi \left[x(\sin x + \cos x) + 2\cos x \right]_0^{\frac{\pi}{4}}$$

$$= 2\pi \left[\frac{\pi}{4} \cdot \frac{2}{\sqrt{2}} + \sqrt{2} - 2 \right] \text{ will do}$$

$$= 2\pi \left(\frac{\sqrt{2}\pi}{4} + \sqrt{2} - 2 \right)$$

(b) (i) Resolving vertically at P,

$$mg + 2T \cos \theta = 3T \cos \theta$$

$$\therefore T \cos \theta = mg$$

(ii) Resolving in direction PO,

$$m\omega^2(OP) = 3T \sin \theta + 2T \sin \theta \quad \text{where } OP = l \sin \theta$$

$$\therefore m\omega^2 l = 5T$$

(iii) From (i), (ii), $\frac{m\omega^2 l}{mg} = \frac{5T}{T \cos \theta}$

$$\therefore \omega^2 l = \frac{5g}{\cos \theta} \quad \text{or} \quad \omega^2 = \frac{5g}{l \cos \theta} > \frac{5g}{l} \quad \text{since } 0 < \cos \theta < 1$$

$$\therefore \omega > \sqrt{\frac{5g}{l}}$$

$$(C) (i) \quad q = 5 + c^2 \quad \therefore c^2 = 4, \quad c = 2$$

$$\therefore S = (2, 0), \quad S'(-2, 0)$$

$$(ii) \quad c = ae \quad \therefore e = \frac{2}{3}$$

$$\text{directrices are } x = \pm \frac{3}{\frac{2}{3}} = \pm \frac{9}{2}$$

$$\therefore PS + PS' = ePD + ePD'$$

$$= e \left(\frac{9}{2} - x_1 + x_1 + \frac{9}{2} \right)$$

$$= 9e$$

Question 6

(a) (i) Any point on $xy=1$ is $(t, \frac{1}{t})$

$$\therefore \text{at } P, Q, R, S \quad t^4 + \frac{1}{t^4} + At + \frac{B}{t} + C = 0$$

$$\text{or } t^4 + At^3 + Ct^2 + Bt + 1 = 0$$

has the roots p, q, r, s

$$\therefore pqrs = 1, \text{ product of roots}$$

$$(ii) \text{ grad } RS = \frac{\frac{1}{r} - \frac{1}{s}}{r - s} = \frac{s - r}{rs(r - s)} = -\frac{1}{rs}$$

$$\therefore \text{grad of } l \text{ is } rs$$

$$\Rightarrow l \text{ is } y = rsx$$

$$(iii) y = \frac{1}{x} \quad \therefore \frac{dy}{dx} = -\frac{1}{x^2} = -\frac{1}{p^2} \text{ at } P$$

$$\therefore \text{tangent at } P \text{ is } y - \frac{1}{p} = -\frac{1}{p^2}(x - p)$$

$$\text{or } p^2y - p = -x + p$$

$$\text{i.e. } x + p^2y = 2p$$

$$(iv) \text{ Tangent at } Q \text{ is } x + q^2y = 2q$$

$$\therefore \text{at } H, (p^2 - q^2)y = 2(p - q)$$

$$\therefore y = \frac{2}{p+q}, \quad x = 2p - p^2 \cdot \frac{2}{p+q} = \frac{2pq}{p+q}$$

$$\text{i.e. } H = \left(\frac{2pq}{p+q}, \frac{2}{p+q} \right)$$

$$\text{If } H \text{ is on } l \text{ then } \frac{2}{p+q} = rs \cdot \frac{2pq}{p+q}$$

$$\text{i.e. } 1 = rs pq \text{ is true from (i)}$$

$$(b) (i) \text{ Put } u_n = \frac{A}{n} + \frac{B}{n+1} + \frac{C}{n+3}$$

$$: A(n+1)(n+3) + Bn(n+3) + Cn(n+1) \equiv 7n+3$$

$$n=0 \Rightarrow 3A=3, \quad A=1$$

$$n=-1 \Rightarrow -2B=-4, \quad B=2$$

$$\therefore 1+2+C=0, \quad C=-3$$

$$\therefore u_n = \frac{1}{n} + \frac{2}{n+1} - \frac{3}{n+3}$$

$$(ii) S_n = \left(1 + \frac{1}{2} + \dots + \frac{1}{n}\right) + 2\left(\frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n+1}\right) - 3\left(\frac{1}{4} + \frac{1}{5} + \dots + \frac{1}{n+3}\right)$$

$$= \left(\frac{1}{4} + \frac{1}{5} + \dots + \frac{1}{n}\right)(1+2-3) + 1 + \frac{1}{2} + \frac{1}{3} + 2\left(\frac{1}{2} + \frac{1}{3} + \frac{1}{n+1}\right) - 3\left(\frac{1}{n+1} + \frac{1}{n+2} + \frac{1}{n+3}\right)$$

$$= 0 + 3\frac{1}{2} - \frac{1}{n+1} - \frac{3}{n+2} - \frac{3}{n+3}$$

$$= \frac{7}{2} - \left(\frac{1}{n+1} + \frac{3}{n+2} + \frac{3}{n+3}\right)$$

$$< \frac{7}{2} \quad \text{since } n=1, 2, \dots$$

Question 7

(a) (i) $m\ddot{x} = mg - mkv$

$$\therefore \ddot{x} = g - kv \Rightarrow g - kU = 0, U = \frac{g}{k}$$

$$\therefore \ddot{x} = k \left(\frac{g}{k} - v \right) = k(U - v)$$

(ii) $m\ddot{x} = -mg - mkv$

$$\therefore \ddot{x} = -k(U + v)$$

$$\therefore v \frac{dv}{dx} = -k(U + v)$$

$$v \frac{dv}{dx} = -k \frac{(U + v)}{v}$$

(iii) From (ii), $\frac{dx}{dv} = -\frac{1}{k} \cdot \frac{v}{U + v}$

$$= -\frac{1}{k} \left(1 - \frac{U}{U + v} \right)$$

$$\therefore [x]_0^H = -\frac{1}{k} \left[v - U \ln(U + v) \right]_{kU}^0$$

$$\therefore H = -\frac{1}{k} (-U \ln U - kU + U \ln U(1+k))$$

$$= -\frac{1}{k} (U \ln(1+k) - kU)$$

$$\text{ie } H = U \left(1 - \frac{1}{k} \ln(1+k) \right)$$

(iv) $\frac{dv}{dt} = k(U - v)$

$$\therefore \frac{dt}{dv} = \frac{1}{k} \cdot \frac{1}{U - v}$$

$$\Rightarrow kt = -[\ln(U - v)]_0^v = -(\ln(U - v) - \ln U)$$

$$\therefore \ln \left(\frac{U - v}{U} \right) = -kt$$

$$\therefore \frac{U - v}{U} = e^{-kt}$$

$$\Rightarrow v = U(1 - e^{-kt})$$

(v) From (iii) $X = \int_0^T U (1 - e^{-kt}) dt$

and $\therefore$ from particle B equation,

$$h - X = \int_0^T k U e^{-kt} - U (1 - e^{-kt}) dt$$

$$= -U [e^{-kt}]_0^T - X$$

$$\therefore h = -U (e^{-kT} - 1)$$

$$\therefore e^{-kT} = \frac{U - h}{U}$$

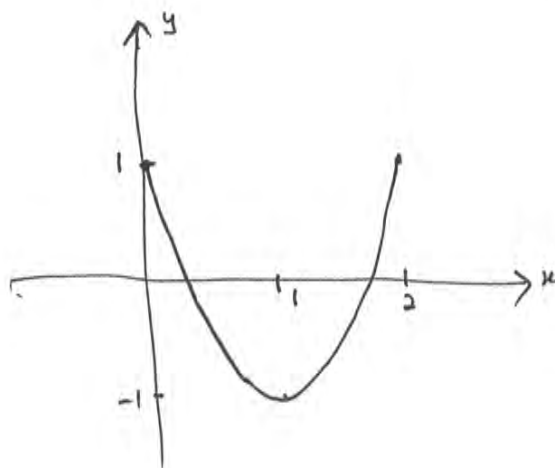
$$\text{or } e^{kT} = \frac{U}{U - h}$$

$$\therefore kT = \ln \left(\frac{U}{U - h} \right)$$

$$\therefore T = \frac{1}{k} \ln \left(\frac{U}{U - h} \right)$$

(b) $y = -\cos 2t = 2\sin^2 t - 1 \Rightarrow y = 2(x-1)^2 - 1$, a parabola

Now $y + 1 = 2(x-1)^2$ where $0 \leq x \leq 2$
and $-1 \leq y \leq 1$



Question 8

(a) (i) Since $2 \leq x \leq 4$, then $\sqrt{2^{2n}-1} < \sqrt{x^{2n}-1}$, $2 < x \leq 4$

$$\Rightarrow \frac{1}{\sqrt{x^{2n}-1}} < \frac{1}{\sqrt{2^{2n}-1}}, \quad 2 < x \leq 4$$

$$\begin{aligned} \therefore J_n &< \int_2^4 \frac{dx}{x \sqrt{2^{2n}-1}} \\ &= \frac{1}{\sqrt{2^{2n}-1}} \int_2^4 \frac{1}{x} dx \\ &= \frac{1}{\sqrt{2^{2n}-1}} \left[\ln x \right]_2^4 = \frac{\ln 2}{\sqrt{2^{2n}-1}} \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad J_n &> \int_2^4 \frac{dx}{x \sqrt{x^{2n}}} \\ &= \int_2^4 \frac{dx}{x^{n+1}} = -\frac{1}{n} \left[\frac{1}{x^n} \right]_2^4 = \frac{1}{n} \left(\frac{1}{2^n} - \frac{1}{4^n} \right) \\ &= \frac{1}{n} \left(\frac{2^n - 1}{4^n} \right) = \frac{2^n - 1}{n \cdot 4^n} \end{aligned}$$

$$\begin{aligned} \text{(iii)} \quad \frac{1}{x \sqrt{x^{2n}-1}} &= \frac{x^{-n-1}}{x^{-n} \sqrt{x^{2n}-1}} = \frac{x^{-n-1}}{\sqrt{x^{-2n}(x^{2n}-1)}} \\ &= \frac{x^{-n-1}}{\sqrt{1-x^{-2n}}} \end{aligned}$$

(iv) From (iii), put $u = x^{-n}$; $x=2, u=2^{-n}$
 $\frac{du}{dx} = -n x^{-n-1}$; $x=4, u=4^{-n}$

$$\begin{aligned} \therefore J_n &= -\frac{1}{n} \int_{2^{-n}}^{4^{-n}} \frac{1}{\sqrt{1-u^2}} du \\ &= \frac{1}{n} \left[\cos^{-1} u \right]_{2^{-n}}^{4^{-n}} = \frac{1}{n} \left(\cos^{-1} 4^{-n} - \cos^{-1} 2^{-n} \right) \end{aligned}$$

or, of course, $-\frac{1}{n} \left(\sin^{-1} 4^{-n} - \sin^{-1} 2^{-n} \right)$

$$(b) \quad (i) \quad u_3 = 2(1+0) = 2$$

$$\therefore u_4 = 3(2+1) = 9$$

$$(ii) \quad A_{n-2} - A_{n-1} = \left(\frac{1}{2!} - \frac{1}{3!} + \dots + (-1)^{n-1} \frac{1}{(n-2)!} \right) - \left(\frac{1}{2!} - \frac{1}{3!} + \dots + (-1)^{n-2} \frac{1}{(n-3)!} + (-1)^{n-1} \frac{1}{(n-1)!} \right)$$

$$= -(-1)^{n-1} \frac{1}{(n-1)!} = (-1)^n \frac{1}{(n-1)!}$$

$$(iii) \quad u_3 = 3! \left(\frac{1}{2!} - \frac{1}{3!} \right) = 3 - 1 = 2$$

$$u_4 = 4! \left(\frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} \right) = 12 - 4 + 1 = 9$$

$$\therefore \text{from (i), } u_n = n! A_n \text{ for } n = 3, 4$$

$$\therefore \text{Assume } u_n = n! A_n \text{ for consecutive integers } n-2, n-1$$

$$\text{Then } u_n = (n-1) (u_{n-1} + u_{n-2})$$

$$= (n-1) \left((n-1)! A_{n-1} + (n-2)! A_{n-2} \right) \text{ using the assumption}$$

$$= (n-1)(n-2)! (n-1) A_{n-1} + A_{n-2}$$

$$= (n-1)! (n A_{n-1} + A_{n-2} - A_{n-1})$$

$$= (n-1)! \left(n A_{n-1} + (-1)^n \frac{1}{(n-1)!} \right) \text{ from (ii)}$$

$$= n! \left(A_{n-1} - (-1)^n \frac{1}{n!} \right) + (-1)^n$$

$$= n! A_{n-1} - (-1)^n + (-1)^n$$

$$= n! A_n$$

$$\therefore \text{by induction } u_n = n! A_n, \quad n \geq 3$$